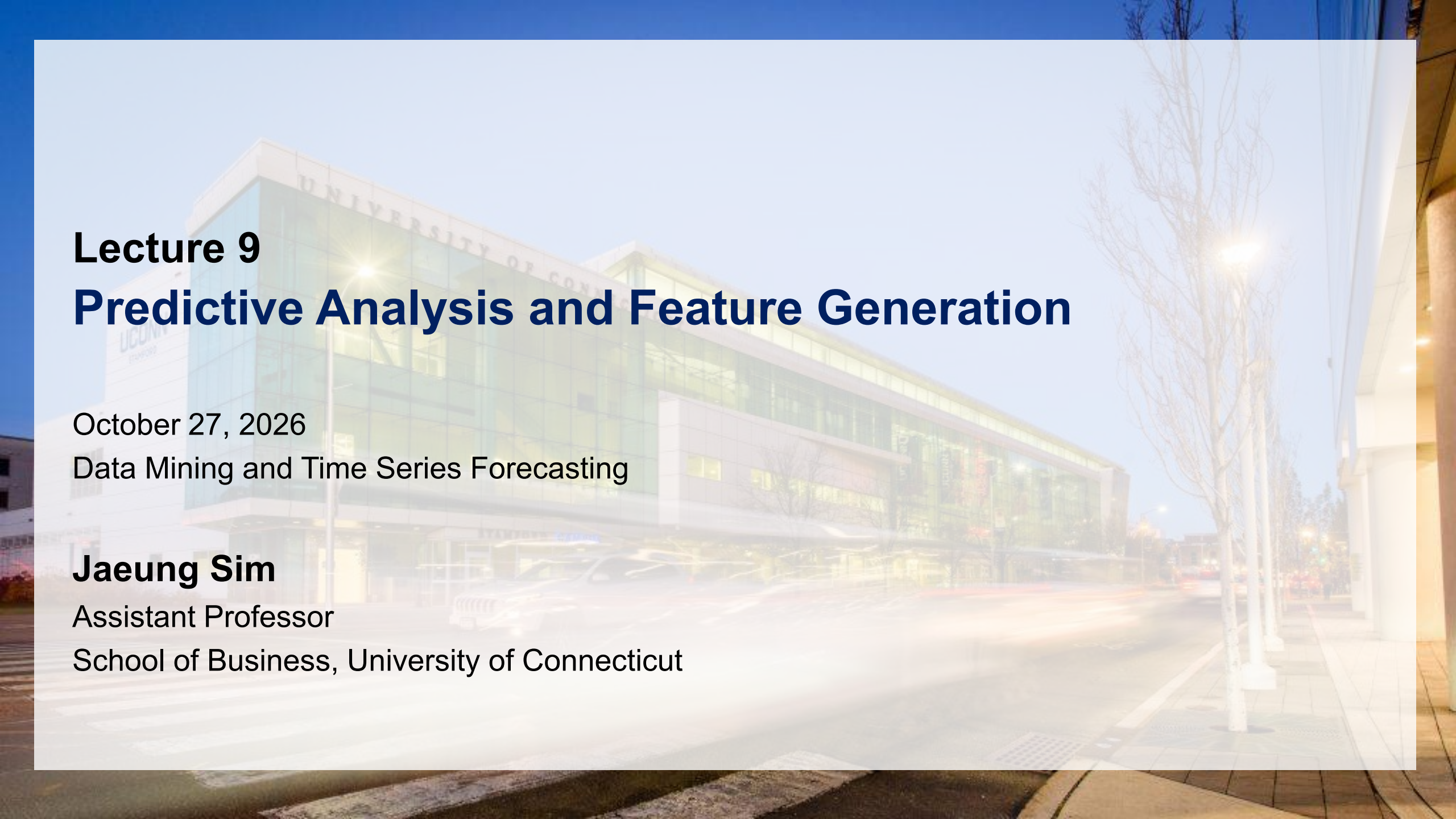


Lecture 9
Predictive Analysis and Feature Generation

October 27, 2026

Data Mining and Time Series Forecasting

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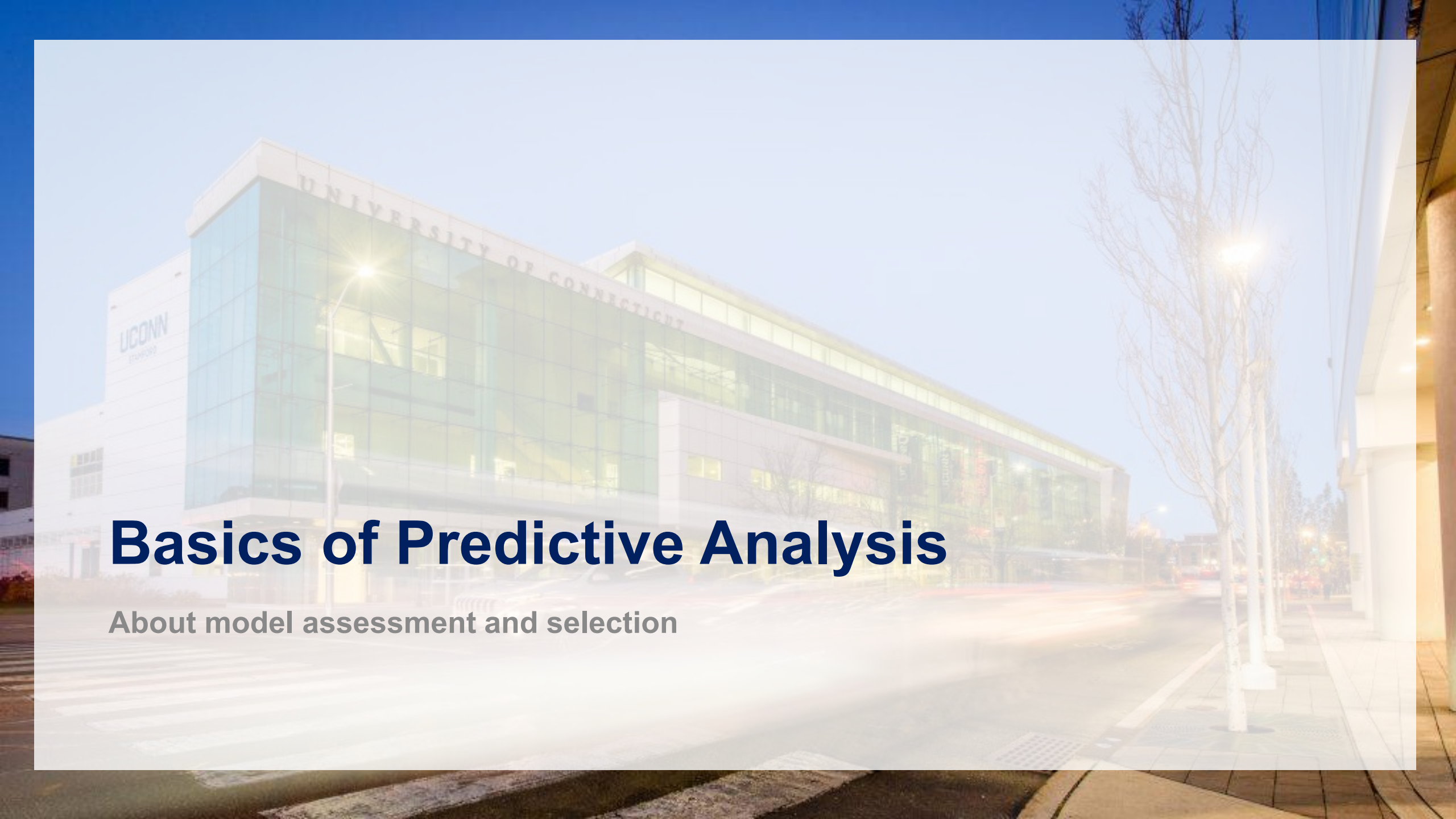
Agenda

- Python Hands-on #1: Text Mining and Singular Value Decomposition
- Basics of Predictive Analysis
- Python Hands-on #2: Predictive Analysis with Feature Generation
- Hands-on Homework #2 Guideline
- Team Project #2 Guideline
- Announcements



Python Hands-on #1

Text Mining and Singular Value Decomposition

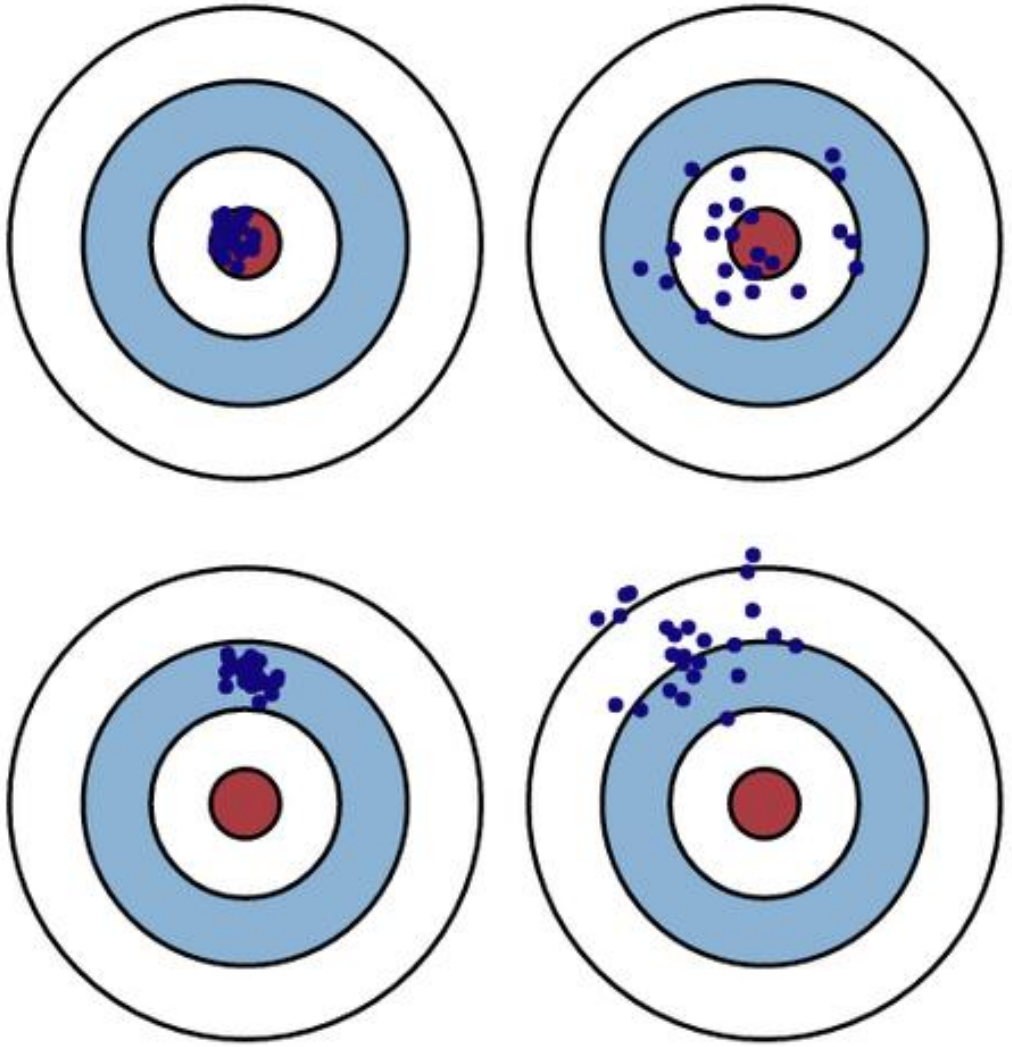


Basics of Predictive Analysis

About model assessment and selection

Bias-Variance Tradeoff

- **Bias vs. Variance**
  - What is a high-bias case?
  - What is a high-variance case?

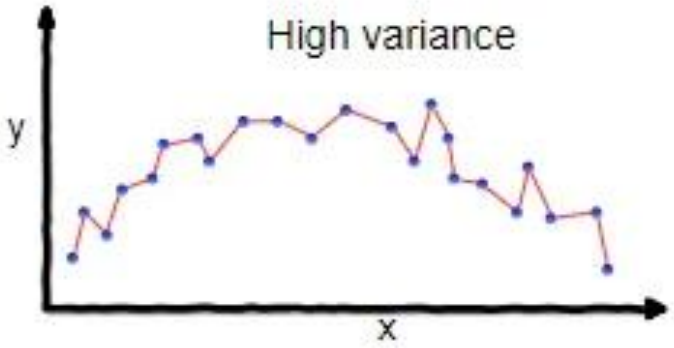


Bias-Variance Tradeoff

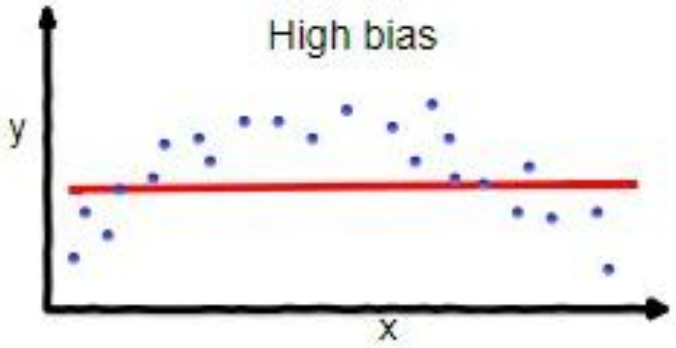
- **Bias vs. Variance in Machine Learning**
  - **Bias** refers to the error introduced by approximating a real-world problem (which may be complex) with a simplified model. High bias means the model is too simple, leading to underfitting—failing to capture the underlying patterns in the data.
  - **Variance** refers to the model's sensitivity to small fluctuations in the training data. High variance means the model is too complex, leading to overfitting—capturing noise along with the signal in the training data.
  - The tradeoff is that reducing bias typically increases variance, and vice versa. The goal is to find the right balance, where the model generalizes well to new, unseen data, without underfitting or overfitting.

Bias-Variance Tradeoff

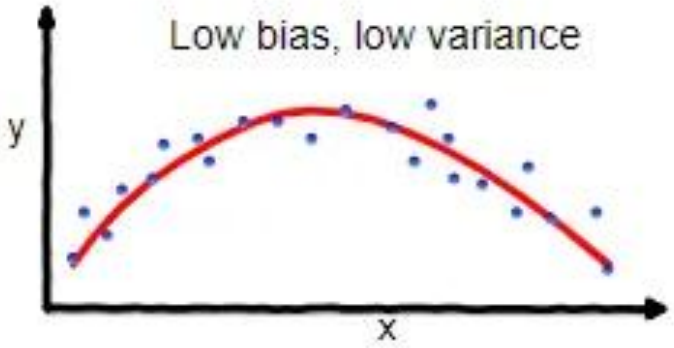
- **Bias vs. Variance in Machine Learning**



overfitting



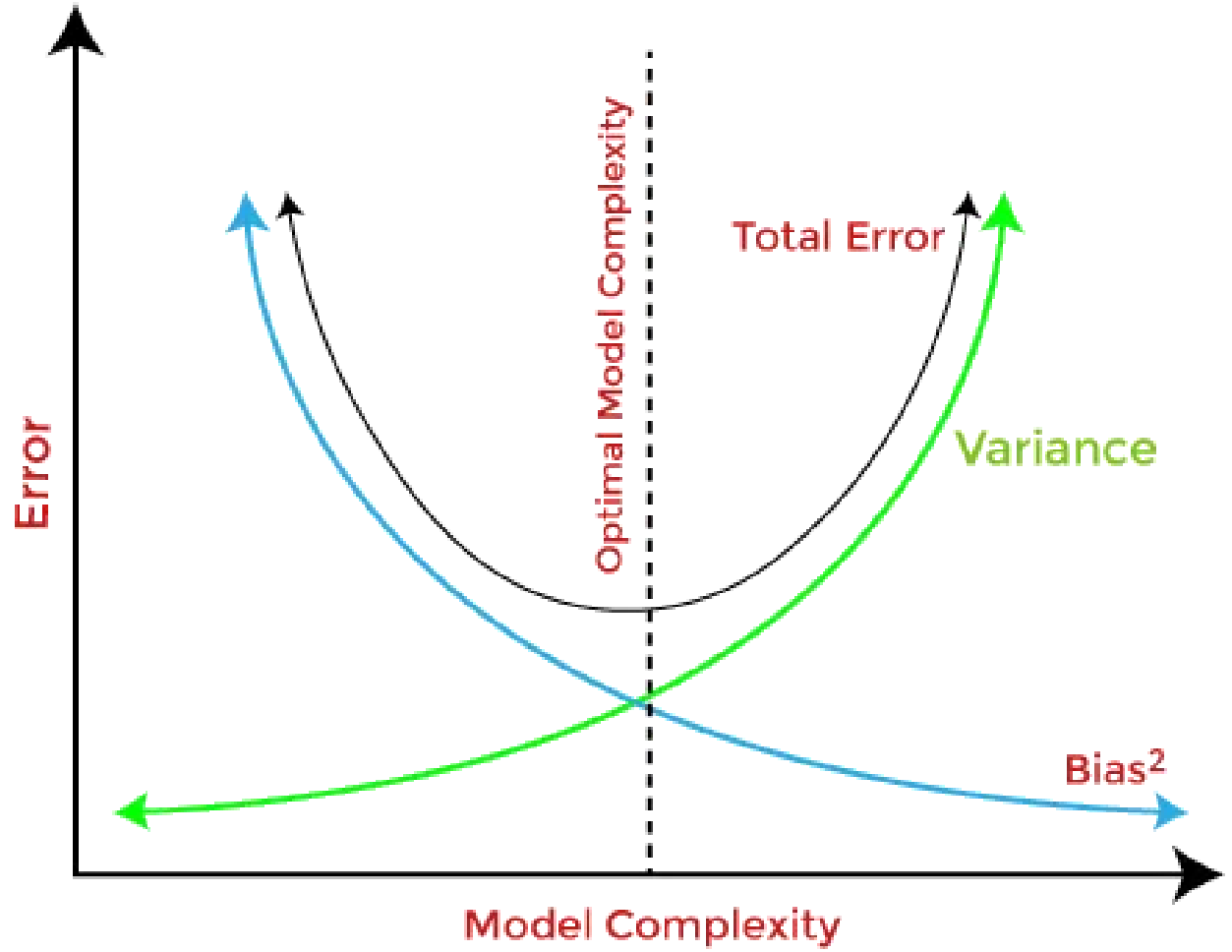
underfitting



Good balance

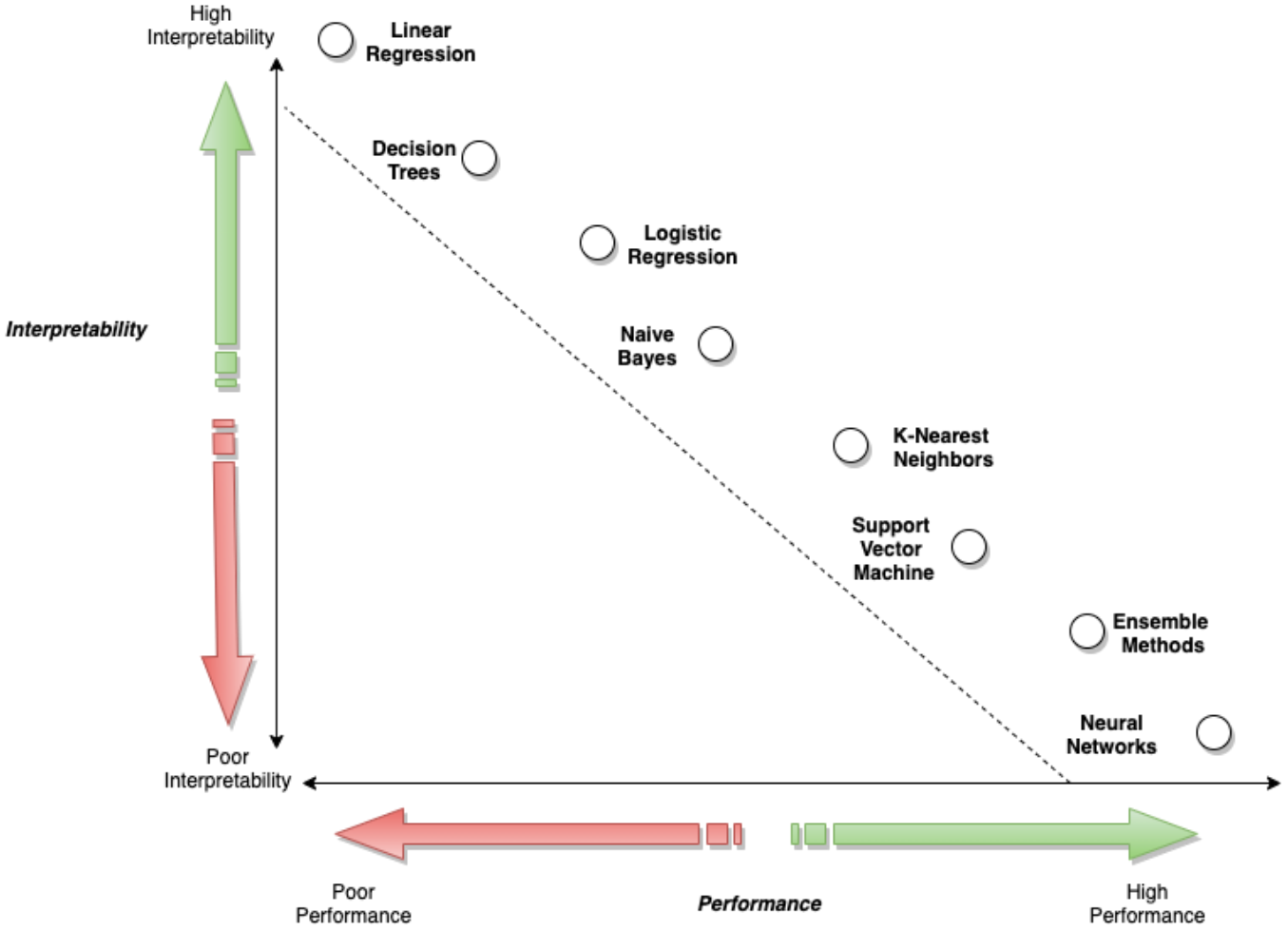
Bias-Variance Tradeoff

- **Bias vs. Variance in Machine Learning**



Performance vs. Interpretability

- **Trade-off by model category**



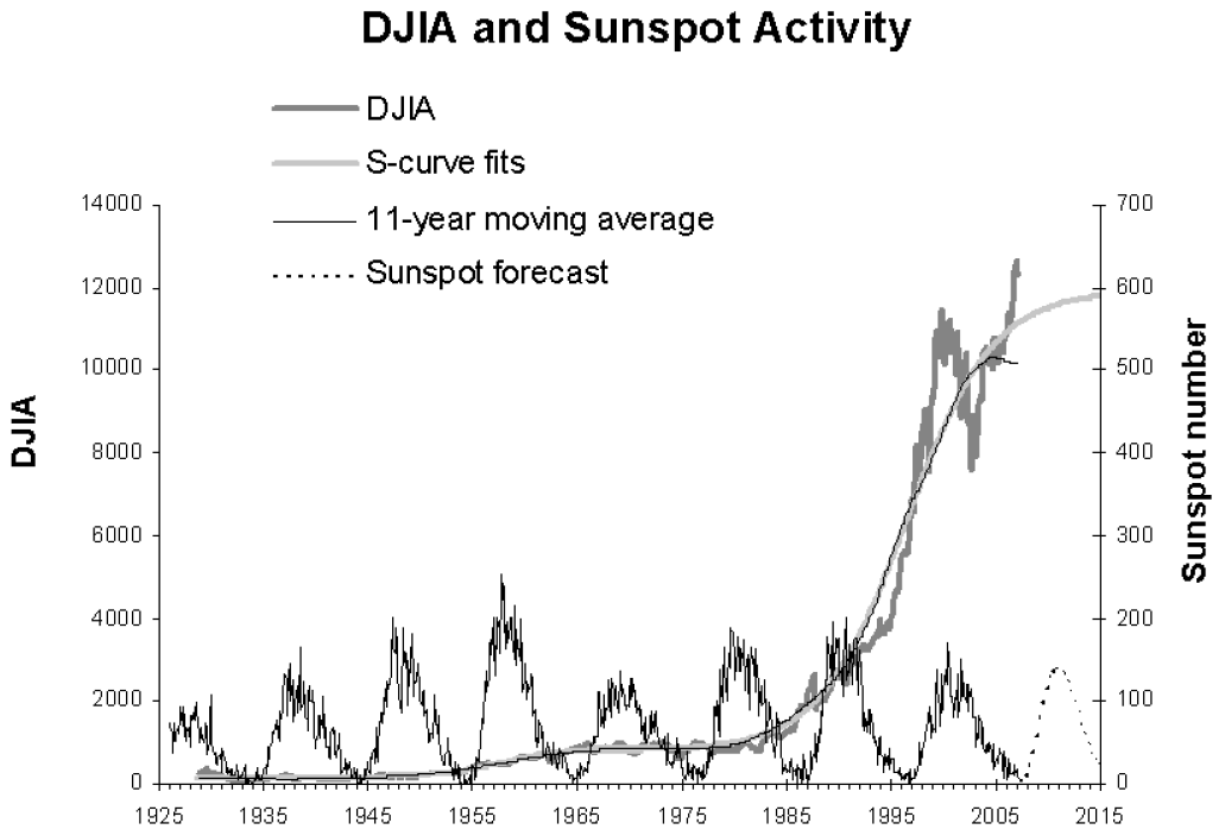
Source: https://docs.aws.amazon.com/whitepapers/latest/model-explainability-aws-ai-ml/interpretability-versus-explainability.html

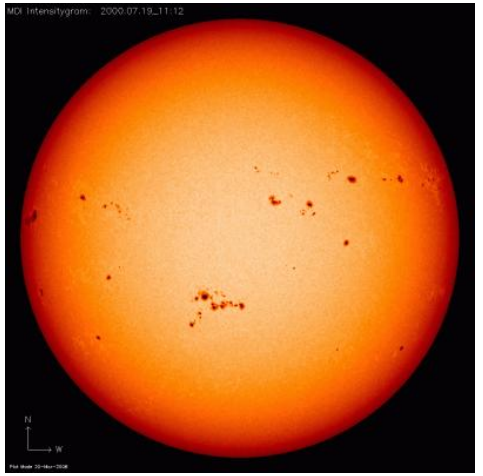
Performance vs. Interpretability

- **Trade-off by variable selection**
  - Some variables can be less interpretable but provide high predictive power.



Dow Jones Industrial Average





Source: Modis, T. (2007). Sunspots, GDP and the stock market. *Technological Forecasting and Social Change*, *74*(8), 1508-1514.

Model Assessment Metrics

- **R-squared** (R^2)
  - **Definition**:
    - R-squared, also known as the **coefficient of determination**, is a statistical measure that represents the proportion of the variance in the dependent variable that is predictable from the independent variables in a regression model.
  - **Formula**: $R^2 = 1 - \frac{\text{Sum of Squared Residuals (SSR)}}{\text{Total Sum of Squares (TSS)}}$
  - **Description**:
    - R-squared measures how well the regression model fits the observed data.
    - Its value ranges from **0 to 1**:
      - **R^2 = 0** means that the model explains none of the variability in the response variable.
      - **R^2 = 1** means the model explains all the variability in the response variable.
  - A higher R-squared value indicates a better fit of the model, but it doesn't necessarily mean the model is good at predicting new data (i.e., it may overfit).

Model Assessment Metrics

- **Adjusted R-squared**
  - **Definition**:
    - Adjusted R-squared modifies the R-squared value to account for the number of predictors in the model. It adjusts for the possibility of overfitting by penalizing the addition of unnecessary variables.
  - **Formula**: $\text{Adjusted } R^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$
    - n denotes number of observations
    - k denotes number of parameters
  - **Description**:
    - Adjusted R-squared accounts for the **number of predictors** in the model.
    - Unlike R-squared, **adjusted R-squared can decrease and be negative** if the added predictors do not improve the model enough to justify their inclusion.
    - In summary, **R-squared** shows how well the model fits the data, while **adjusted R-squared** accounts for the complexity of the model, preventing overfitting by penalizing unnecessary predictors.

Model Assessment Metrics

- **Akaike Information Criteria (AIC)**
  - **Definition:**
    - The Akaike information criterion (AIC) is an estimator of prediction error and thereby relative quality of statistical models for a given set of data.
  - **Formula:** $\mathrm{AIC} = 2k - 2\ln(\hat{L})$
    - k: the number of parameters
    - $\hat{L}$: the maximized value of the likelihood function
  - **Description:**
    - Given a set of candidate models for the data, the preferred model is the one with the **minimum AIC value**.
    - Thus, AIC rewards the likelihood function, but it also includes a **penalty** that is an increasing function of **the number of estimated parameters**.

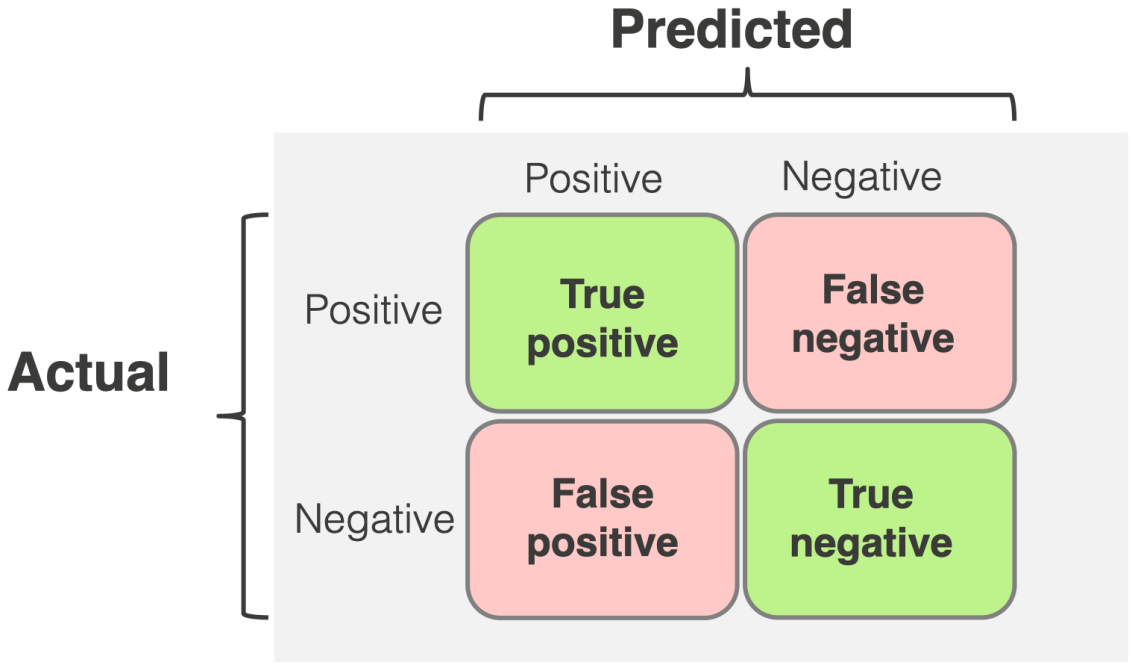
Model Assessment Metrics

- **Bayesian Information Criteria (BIC)**
  - **Definition:**
    - The Bayesian information criterion (BIC) or Schwarz information criterion (also SIC, SBC, SBIC) is a criterion for model selection.
  - **Formula:** $\mathrm{BIC} = kln(n) - 2\ln(\hat{L})$
    - k: the number of parameters
    - $\hat{L}$: the maximized value of the likelihood function
  - **Description:**
    - Similarly to AIC, models with lower BIC are generally preferred, and BIC penalizes the number of estimated parameters.
    - With AIC the penalty is $2k$, whereas with BIC the penalty is $kln(n)$. Therefore, **BIC penalizes the model's complexity more severely than AIC**.

Model Assessment Metrics

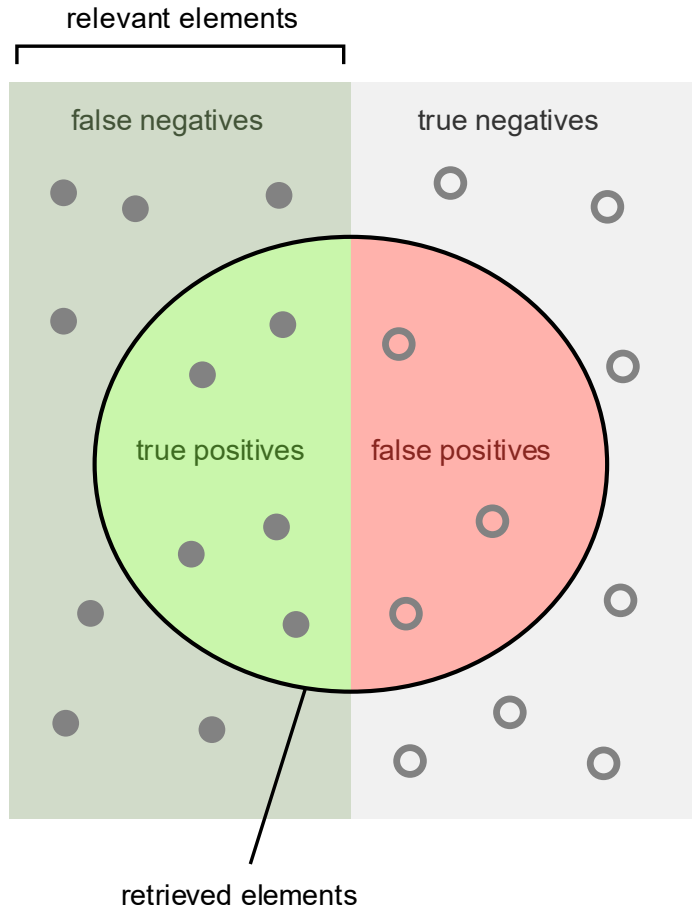
- **Confusion Matrix**
  - **Precision:** the percentage of selected documents that are correctly classified
  - **Recall:** the percentage of all documents in the requested category that are selected
  - **F1 score:** the harmonic mean of precision and recall
    - $\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$
    - An F1 score is considered perfect when it's 1, while the model is a total failure when it's 0
  - **Misclassification Rate:** the percentage of incorrectly classified documents

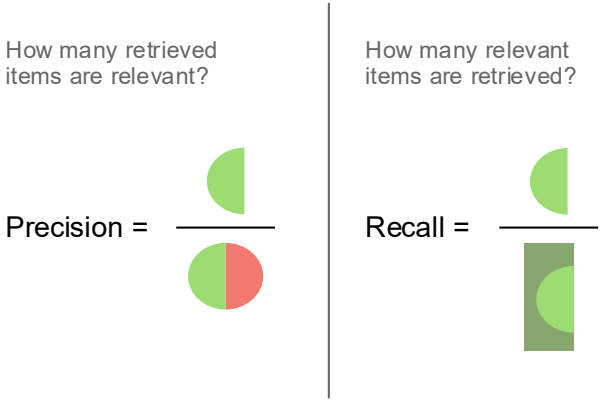
In the confusion matrix, **"actual"** and **"true"** are different!



Model Assessment Metrics

- **Confusion Matrix**
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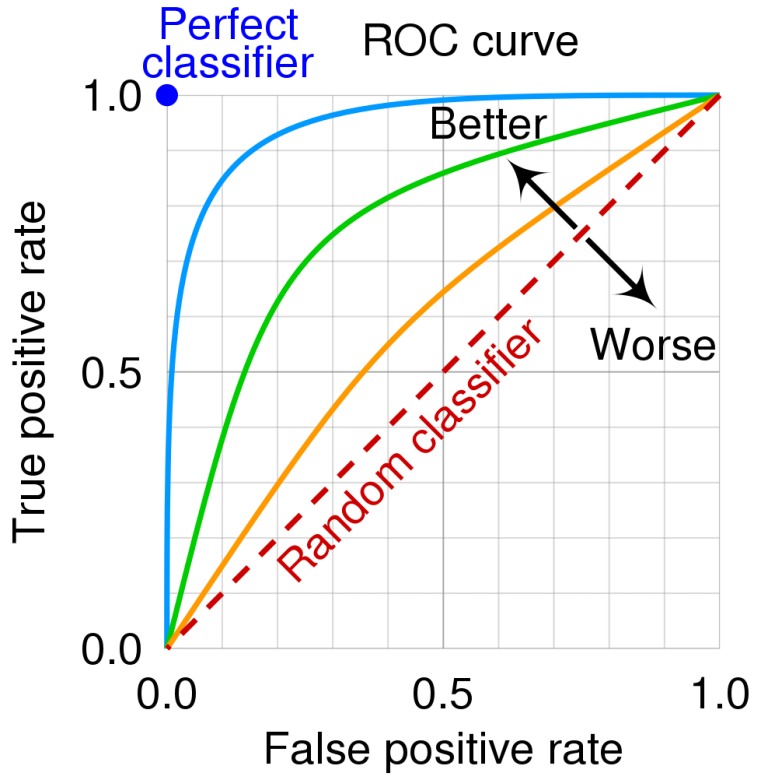
Model Assessment Metrics

- **Confusion Matrix (Example):**
  - Here is a COVID-19 test kit developed by an amateur engineer. It is shown that when you're actually positive, the test result will be positive with 95% possibility. When you're actually negative, the test result will be positive with 5% possibility.
  - Currently, the actual positive rate of COVID-19 in this sample is 5%.
  - Calculate the expected values for the sample of 2,000 people.
    - Precision: 50%
    - Recall: 95%
    - The number of actual positive: 100
    - The number of true positive: 95
    - The number of false positive: 95

N = 2,000		Predicted	
		Positive	Negative
Actual	Positive	95	5
	Negative	95	1805

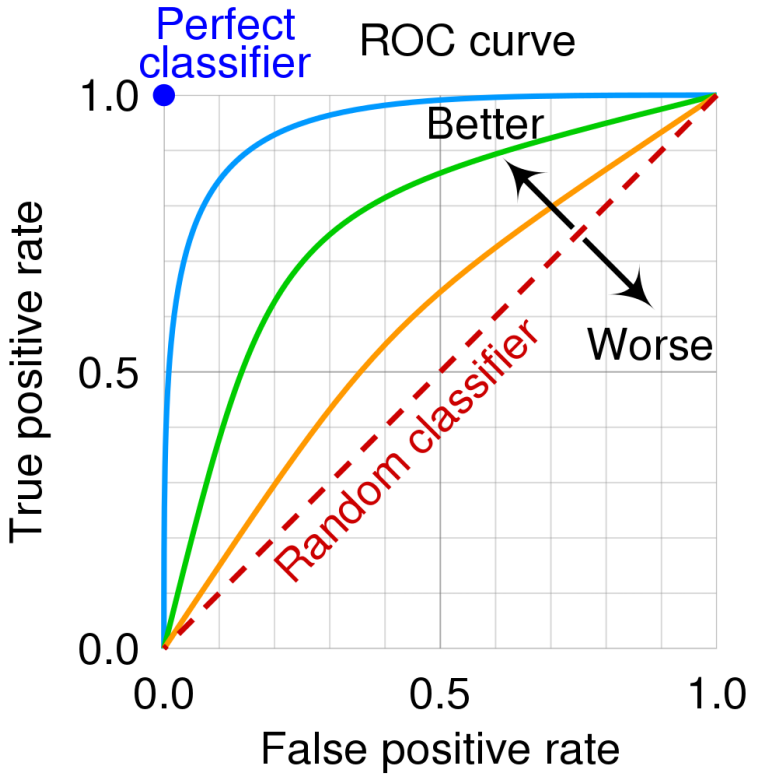
Model Assessment Metrics

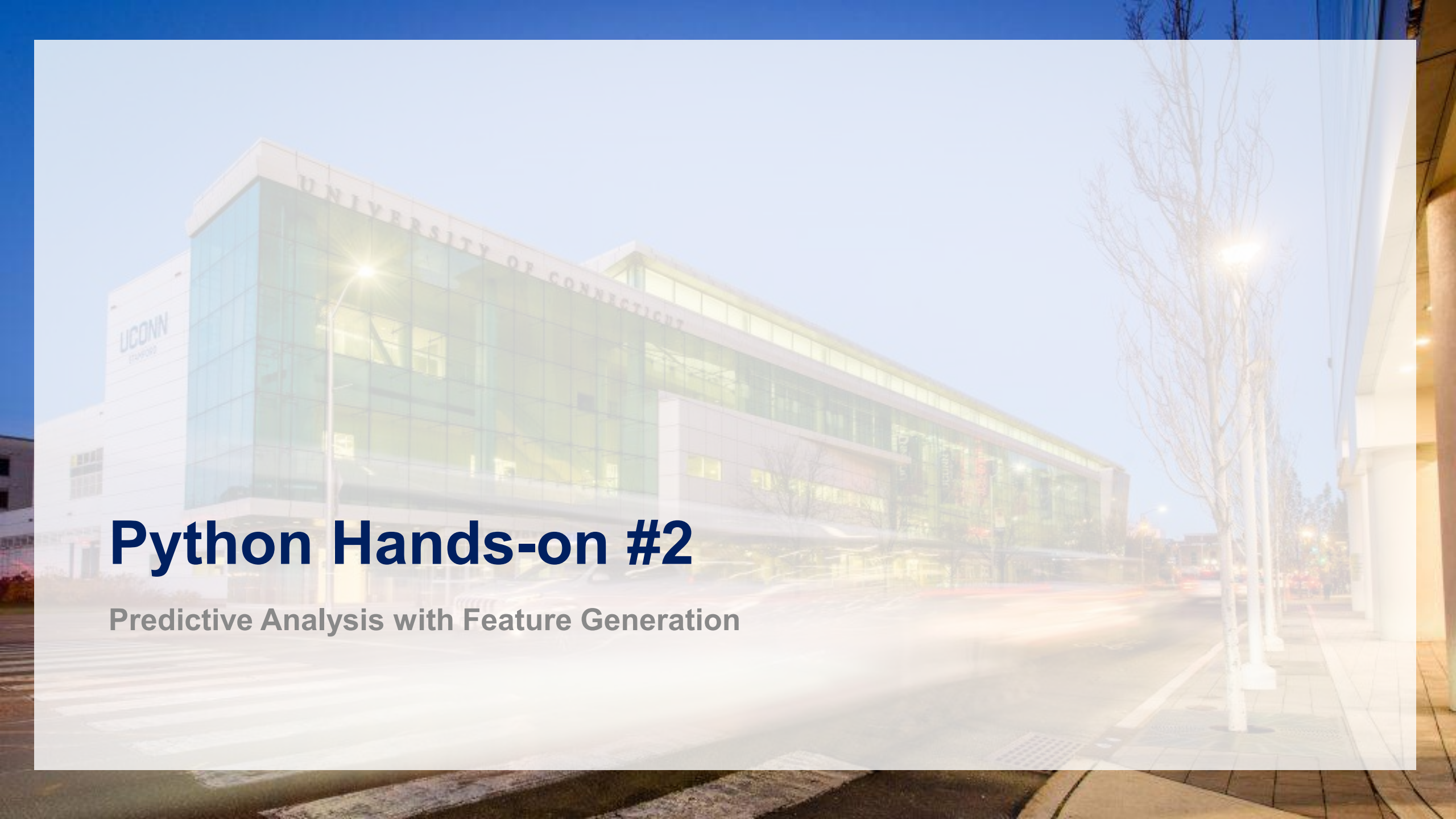
- **ROC Curve**
  - **R**eceiver **O**perating **C**haracteristic Curve
    - A graph showing the performance of a classification model at all classification thresholds
    - An ROC curve plots TPR vs. FPR at different classification thresholds where
      - $TPR = \frac{TP}{TP+FN}$ (true positive rate = recall)
        - True positive among actual positive
      - $FPR = \frac{FP}{FP+TN}$ (false positive rate)
        - False positive among actual negative
    - As you predict more observations as positive, you will catch more actual positives (higher recall) but misclassify negative cases as positive.



Model Assessment Metrics

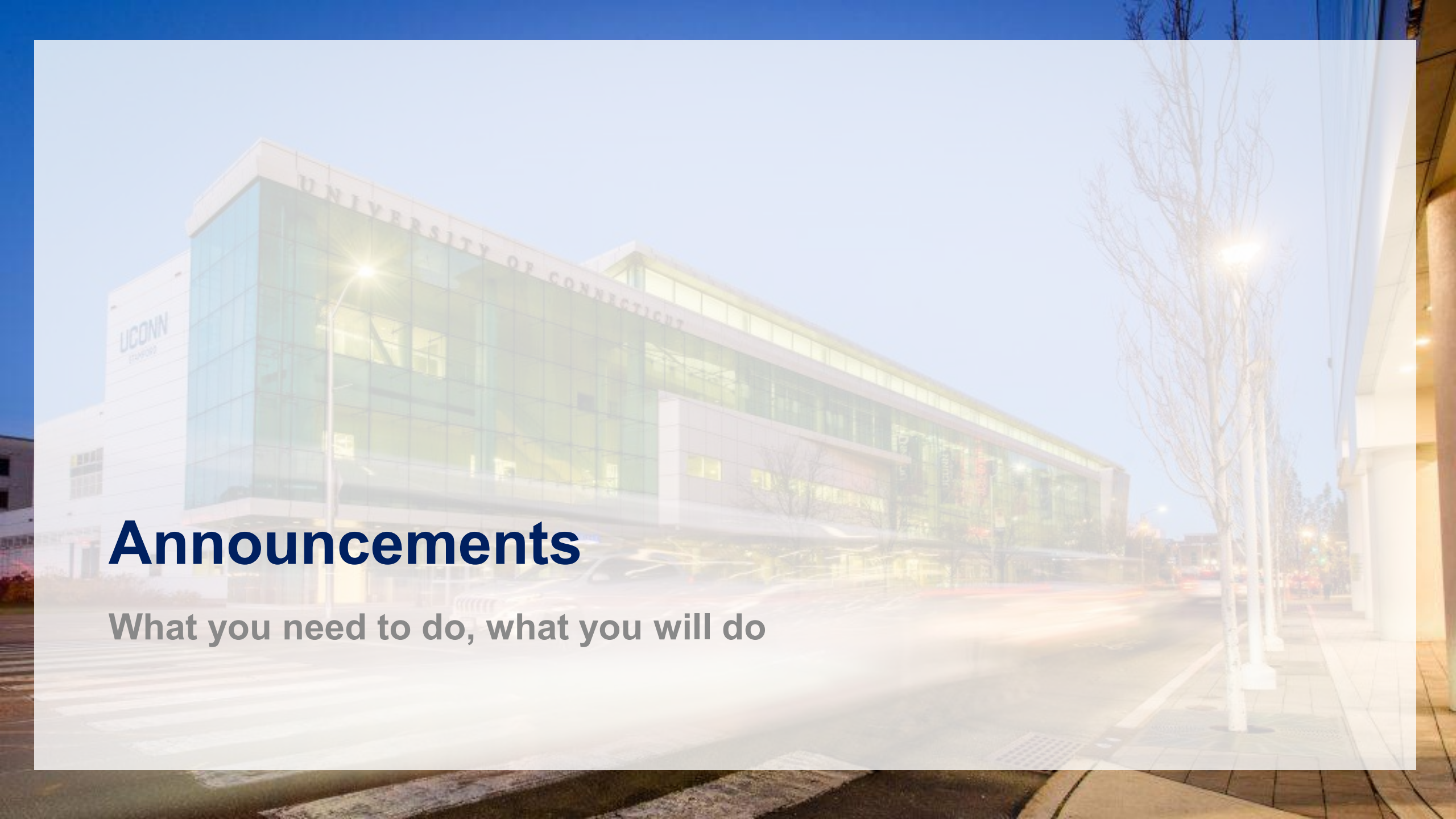
- **ROC Curve**
  - **AUC (Area Under the ROC Curve)**
    - Provides an aggregate measure of performance across all possible classification thresholds.
    - An AUC value of 0.5 suggests no discrimination, meaning the model's ability to distinguish between positive and negative classes is no better than random guessing.
    - An AUC value of 1.0 means the model has perfect discrimination, where it can perfectly differentiate between the positive and negative classes.
    - An AUC value of 0.0 would mean the model is perfectly wrong, but in practice, a model with an AUC of 0 can simply be inverted to achieve an AUC of 1.





Python Hands-on #2

Predictive Analysis with Feature Generation



Announcements

What you need to do, what you will do

Announcements

- **Nov 2 (Mon), 2026: Team Project #2 Proposal Deadline**
- **Nov 3 (Tue), 2026: Deep Learning and Text Mining**
  - **Take-home Midterm results will be announced.**
  - **Nov 8 (Sun), 2026: Homework #2 Deadline**
- **Nov 10 (Tue), 2026: Recent Challenges in Machine Learning**
- **Nov 17 (Tue), 2026 (Virtual): Term Project Presentation #2**
- **Nov 24 (Tue), 2026 (No Class): *Thanksgiving Recess***
- **Dec 1 (Tue), 2026 (Asynchronous): Take-home final exam will be released.**



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